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Bernoulli's Trials & Distribution.

$$p(x_j) = \begin{cases} p & x_j = 1, \; j = 1, 2, \dots n. \\ 1 - p = q & x_j = 0, \; j = 1, 2, \dots n. \\ 0 & \text{otherwise.} \end{cases}$$

$E(x_j) = p$

$V(x_j) = p(1-p)$

Binomial Distribution:

The random variable x that denotes the number of successes (p) in n Bernoulli Trials has a binomial distribution given by

$$P(x) = \begin{cases} \binom{n}{x} p^x q^{n-x} & x = 0, \dots, n. \\ 0, & \text{otherwise} \end{cases}$$

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$E(x) = p + p + p + \dots + p = np.$

$V(x) = pq + pq + pq + \dots + pq = n$

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6th Aug.

Q A production process manufactures computer chips on an average at 2% non-conforming. Everyday a sample size of 50 is taken, if there are more than two non-confirming chips, then the process will be stopped. Compute the probability that the process is stopped by this scheme.

Ans
$$P(x) = \begin{cases} \binom{50}{x} (0.02)^x (0.98)^{50-x} & 0 \le x \le 50 \\ 0 & \text{otherwise.} \end{cases}$$

$P(x \le 2)$

$= P(x=0) + P(x=1) + P(x=2)$

$= \sum_{x=0}^{2} \binom{50}{x} (0.02)^x (0.98)^{50-x}$

$P(x=0) = {}^{50}C_0 (0.02)^0 (0.98)^{50}$

$= 1 \cdot 1 \cdot (0.98)^{50}$

$= (0.98)^{50} = 0.3641$

$P(x=1) = {}^{50}C_1 (0.02)^1 (0.98)^{49}$

$= 50 \times 0.02 \times (0.98)^{49}$

$= 0.371$

$P(x=2) = {}^{50}C_2 \times (0.02)^2 (0.98)^{48}$

$= 0.371 \quad 0.1858$

$\therefore P(x \le 2) = 0.92$

$\therefore P(x > 2) = 1 - 0.92 = 0.08$

$E(x) = np$
$= 50 \times 0.02$
$= \underline{1}$

$\therefore V(x) = npq$
$= 50 \times 0.02 \times 0.98$
$= \underline{0.98}$

Q A recent survey indicated, 82% of single women aged 25 years old will be married in their lifetime. Using Binomial Distribution, find the probability of 2 or 3 women in a sample of 20 will never be married.

Ans: $n = 20$.
$p = 0.18$
$q = 0.82$.

$$P(x) = \begin{cases} \binom{n}{x} p^x q^{n-x} & 0 \le x \le \cancel{20}\, n \\ 0 & \text{otherwise.} \end{cases}$$

$\therefore P(x=2) = {}^{20}C_2 \cdot (0.18)^2 \cdot (0.82)^{18}$
$= 0.1729$

$\therefore P(x=3) = {}^{20}C_3 \, (0.18)^3 \, (0.82)^{17}$

$= 0.227$

$P(2) + P(3) = 0.399999 = \underline{0.4}$

$E(x) = np = 3.6$

$\therefore$ Ans $= 40\%$.

$V(x) = npq = 2.952$

Q The team is currently winning 0.55 of their games. There are 5 games in next 2 weeks. Find the probability that they win more games than they loose (go. for Binomial Distribution)

$n = 5$.
$p = 0.55$
$q = 0.45$.

$\therefore P(x=3) = {}^5C_3 \cdot (0.55)^3 \, (0.45)^2$
$= 0.3369$

$P(x=4) = {}^5C_4 \, (0.55)^4 \, (0.45)^1$
$= 0.205$.

$P(x=5) = {}^5C_5 \cdot (0.55)^5 \, (0.45)^0$
$= 0.0503$

$P(x) = P(3) + P(4) + P(5)$

$\therefore P(x) = 0.592$

$= 59.2\%$

$E(x) = np = 2.75$

$V(x) = npq = 1.2375$.

Geometric & Negative Distribution.

geometric:

$$P(x) = \begin{cases} q^{x-1} p, & x = 1, 2, \ldots \\ 0 & \text{otherwise} \end{cases}$$

$$E(x) = \frac{1}{p}$$

$$V(x) = \frac{q}{p^2}$$

:) geometric distribution is related to the sequence of Bernoulli's trials where random variable of interest x is defined to be number of trials to achieve the 1st success.

Negative:

$$P(y) = \begin{cases} \binom{y-1}{k-1} q^{y-k} p^k, & y = k, k+1, \ldots \\ 0 & \text{otherwise} \end{cases}$$

$$E(y) = k/p$$

$$V(y) = \frac{kq}{p^2}$$

Negative Binomial distribution is for the number of trials until the k^{th} success [for $k = 1, 2, 3, \ldots, n$]

Q 40% of assembled printers are rejected at the inspection station. Find the probability that 1st acceptable printer is the 3rd one inspected.

$p = 0.60$

$q = 0.4$

$x = 3$.

$P(x) = (0.4)^2 \; 0.6$

$= 0.16 \times 0.6$.

Q Joe is 3^{rd} string quarterback at the university. the probabil that joe gets into any game is 0.4 find the probability of

i> 1^{st} game joe enters is the 4^{th} games of the season.

ii> joe plays no more than two of the 1^{st} five games.

i> $p = 0.4$

$q = 0.6.$

$x = 4.$

$\therefore \; P(x=4) = (0.6)^3 (0.4)$

$= 0.216 \times 0.4$

$= \underline{0.0864}$

ii> $p = 0.6$ (Binomial Distribution)

$q = 0.4$ Joe not playing

$n = 5$

$P(x \leq 2)$

$P(0) = {}^5C_0 \cdot (0.4)^0 (0.6)^5$

$= 1 \times 1 \times 0.07776 = 0.07776$

$P(1) = {}^5C_1 \; (0.4)^1 (0.6)^4$

$= 0.2592$

$P(2) = {}^5C_2 \; (0.4)^2 (0.6)^3$

$= 0.3456$

$= \underline{0.68256}$

9th Aug 2010.

Poisson Distribution

~~pdf~~ pmf $\quad p(x) = \begin{cases} \dfrac{e^{-\alpha} \alpha^x}{x!}, & x = 0, 1, \ldots \\ 0, & \text{otherwise} \end{cases}$

where $\alpha > 0$

$E(x) = \alpha = V(x)$

$\sigma = \sqrt{V(x)} = \sqrt{\alpha}$

cdf

$F(x) = \sum_{i=0}^{x} \frac{e^{-\alpha} \alpha^i}{i!}$

$\cdot$) Poisson distribution describes many random processes & mathematically finds it

Q The number of calls that a computer repair person gets are in accordance with a poisson distribution with a mean of $\alpha = 2$ per hour. Find the probability of 3 calls in the next hour.

Random => Poisson

Q Records indicate 1.8 % of students entering the university, drop out by midterm. Find probability of three or fewer students will drop out of random group of 200 students.

for 200 students

$\alpha = \underline{1.8\%} = \frac{1.8}{100} \times 200$

$\alpha = \underline{\underline{3.6}}$

$\therefore P(x=0) = \frac{e^{-3.6} 3.6^0}{0!} = 0.027$

$\therefore P(x=1) = \frac{e^{-3.6} 3.6^1}{1!} = 0.09$

$\therefore P(x=2) = \frac{e^{-3.6} 3.6^2}{2} = 0.17$

$\therefore P(x=3) = \frac{e^{-3.6} 3.6^3}{3 \times 2} = 0.21$

$= \underline{0.515}$

$\therefore P(x \leq 3) = 0.5152$

Q Lane is a popular student. He recieves on an average 4 calls a night.

→ What is the probability that tomorrow the calls received will exceed the average by more than one standard deviation?

Standard Deviation $\sigma = 2$.

$\therefore V(x) = \alpha = \sigma^2 = 2^2 = 4$

Calls should exceed by one standard deviation.

$\therefore$ more than 6 calls.

$\therefore P(x > 6) = ?$

$\therefore F(x) = \sum_{i=0}^{x} \frac{e^{-\alpha} \alpha^i}{i!}$

$= e^{-4} \sum_{i=0}^{x} \frac{\alpha^i}{i!}$

$= e^{-4} \left[\frac{4^0}{1} + \frac{4^1}{1} + \frac{4^2}{2} + \frac{4^3}{6} + \frac{4^4}{24} + \frac{4^5}{120} + \frac{4^6}{720} \right]$

$= 0.8893$

$\therefore P(x > 6) = 1 - 0.8893 = 0.1106$

Q Poission distribution can be used to approximate binomial distribution when n is large & p is small

n ↑ no. of sample. p ↑ probability

→ poission mean (λ)

$\alpha = np$

When n is very large & p is very small the two distribution converge.

Q In production of Bearing depression occur making it unfit for sale. On average one in every 800 of the bearings has defects. What is the probability that a random sample of 4000 bearing will yield fewer than 3 depressions.

$3 \to \lambda$

Binomial Distribution.

$p = 1/800 \qquad q = 799/800$

$n = 4000$

$p(x) = \binom{n}{x} p^x q^{n-x}$

$$P(x)_{x<3} = {}^{4000}C_0 \left(\frac{1}{800}\right)^0 \left(\frac{799}{8000}\right)^{4000} + {}^{4000}C_1 \left(\frac{1}{800}\right)^1 \left(\frac{799}{800}\right)^{3999} + {}^{4000}C_2 \left(\frac{1}{800}\right)^2 \left(\frac{799}{800}\right)$$

$= 0.006717 + 0.03363$

0.08415

$= 0.12449$

$\therefore P(x<3) = \underline{0.1245}$

Poission Distribution.

$\therefore \alpha = 4000 \times \frac{1}{800} = n \times p$

$\alpha = np$

$\alpha = 5$

$$\therefore P(x)_{x<3} = \sum_{i=0}^{x} \frac{e^{-\alpha}\alpha^x}{x!}$$

$$= e^{-5}\left[\frac{5^0}{1} + \frac{5^1}{1} + \frac{5^2}{2}\right]$$

$= \underline{0.1246}$

$\therefore P(x<3) = \underline{0.1246}$

Q A car service station receives cars @ 5 cars per hour in accordance with poission. Find probability that a car will arrive two hours after its predecessor.

$\alpha = 5 \times 2 = 10.$

↑ because two hours.

$$P(x=0) = \frac{e^{-10} \cdot 10^0}{0!}$$

Inbetween 2 hrs no car arrives

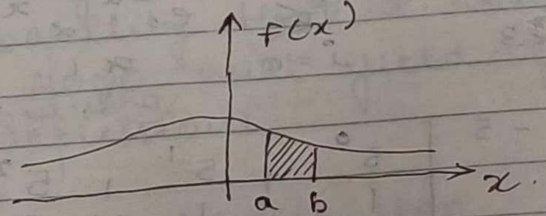
$$= e^{-10}$$

$$=$$

$$= \underline{\underline{0.0000453992976}}$$

⟹ Continuous Random Variables.

$$P(a \le X \le b) = \int_a^b f(x)\,dx$$



$$\text{cdf} \Rightarrow F(x) = \int_{-\infty}^{x} f(t)\,dt$$

Q The life of a device used to inspect cracks in the air craft wings given by x has the following pdf.

$$f(x) = \begin{cases} \frac{1}{2} e^{-x/2}, & x \ge 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the probability that the life of the device is between 2 to 3 years.

$$P(2 \le X \le 3) = \frac{1}{2}\int_2^3 e^{-x/2}\,dx$$

$$= \frac{1}{2\,(-1/2)} \left[e^{-x/2} \right]_2^3$$

$$= -\left[e^{-3/2} - e^{-2/2} \right]$$

$$= e^{-1} - e^{-3/2}$$

$$= \underline{0.1447}$$

Prob. that device will last for < 2 yrs,

$$P(0 \le X \le 2) = F(2) - F(0) = F(2) = \frac{1}{2}\int_0^2 e^{-x/2}\,dt = 0.632$$

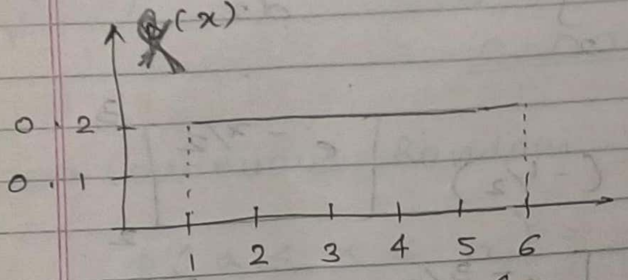
<u>Uniform Distribution:</u>

$$\text{pdf } p(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$

$$\text{cdf } F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \le x < b \\ 1, & x \ge b \end{cases}$$

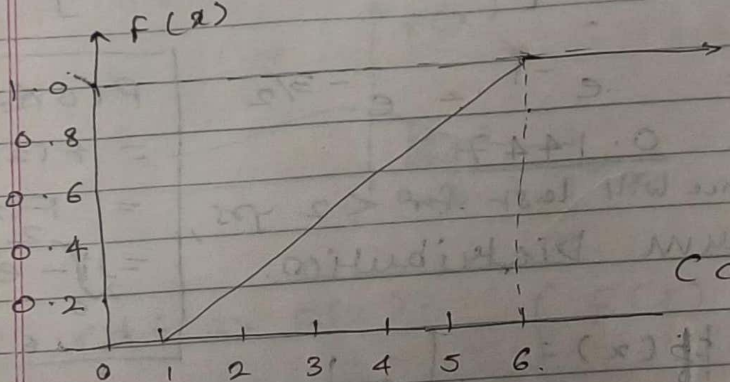
$$E(x) = \frac{a+b}{2}$$

$$V(x) = \frac{(b-a)^2}{12}$$



(pdF).

$a = 1,\ b = 6.$



(cdF)

A random variable x is uniform distributed over $[a, b]$ if its pdf is given by $f(x)$.

Q. A bus arrives every twenty minutes at a stop begining at 6.40 am & continuing till 8.40 am. A person does not know the schedule but arrives randomly between 7 to 7.30 am in the morning.

What is the probability that he wait for more than 5 minutes for the bus.

6.40
7.00
7.20 —— 7.15
7.40
8.00
8.20
8.40

$X \Rightarrow$ Random variable that denotes no. of minutes past 7 that person arrives.

$P(0 < X < 15)$
$+ P(20 < X < 30)$

$$= F(15) - F(0) + F(30) - F(20).$$

$$=$$

$$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{x-b}, & a \le x < b \\ 1, & x \ge b. \end{cases}$$

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$a = 0, \ b = 30.$

$$= \frac{15-0}{30-0} + 1 - \frac{20-0}{30-0}$$

$$= \frac{1}{2} + 1 - \frac{2}{3}$$

$$= \frac{3}{2} - \frac{2}{3}$$

$$= \frac{9-4}{6} = \frac{5}{6} = 0.83$$

13th August 2010.

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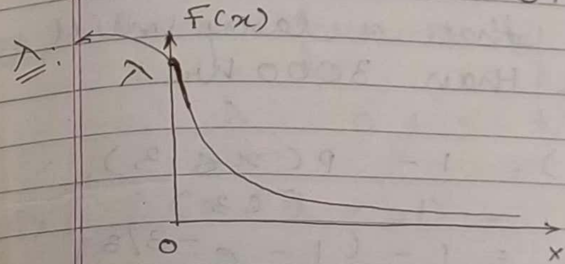
Exponential Distribution

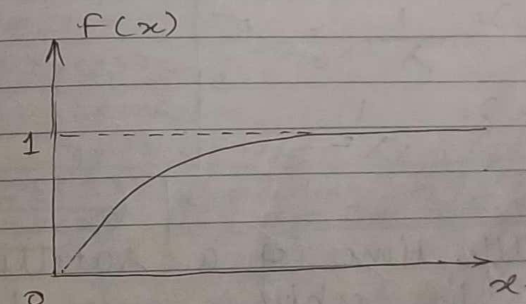
Called as Memoryless Distribution

A random variable 'x' is said to be exponentially distributed with parameter $\lambda > 0$ if the pdf is

$$\text{pdf} \Rightarrow f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$



$$\text{cdF} = \begin{cases} 0, & x < 0 \\ \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}, & x \geq 0 \end{cases}$$



Mean life $\Rightarrow E(x) = \frac{1}{\lambda} = \sigma$ | $V(x) = \frac{1}{\lambda^2}$

Q Life of an Industrial Lamp in 1000 hours is exponentially distributed with failure rate $\lambda = 1/3$. Find the probability that the lamp will last longer than its mean life.

Ans: Failure of lamp is expected after 3000 hrs.

$E(x) = 1/(1/3) = 3 \Rightarrow$ in thousand hours.

Probability that a lamp will last more than 3000 hrs.

$$P(x > 3) = 1 - P(x \leq 3)$$
$$= 1 - F(3)$$
$$= 1 - (1 - e^{-3/3})$$
$$= 1 - (1 - e^{-3/3})$$
$$= 1 - 1 + e^{-1}$$
$$= e^{-1}$$
$$= \underline{0.36787}$$

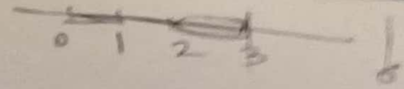
$E(x) = \frac{1}{\lambda} = \sigma$

$V(x) = \frac{1}{\lambda^2}$

lamp will last between (2000 & 3000 hrs)
$= F(3) - F(2)$
$= -0.368 + 0.513$
$= 0.145$

memoryless property | lamp will last another 1000 hrs given it operates 2500 | $P\left(\frac{x > 3.5}{x > 2.5}\right) = e^{-1/3} = 0.71$ $P(x>1)$

Q The life time of a satellite placed in orbit is given by



pdf

$$f(x) = \begin{cases} 0.4 e^{-0.4x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

What is the probability that the

i) satellite is still alive after 5 years.

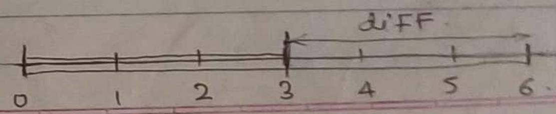
ii) satellite dies in 3 to 6 years from the time it is placed in orbit

Ans: From formula of pdf

$\therefore \lambda = 0.4 = 4/10 = 2/5$

i) $\therefore P(x > 5) = 1 - P(x \leq 5)$ $\quad 1 - F(5)$
$$= 1 - (1 - e^{-2/5 \times 5})$$
$$= 1 - (1 - e^{-2})$$
$$= 1 - 1 + e^{-2}$$
$$= \underline{0.13533}$$

ii) $\therefore P(3 \leq x \leq 6) = F(6) - F(3)$
$$= (1 - e^{-2/5 \times 6}) - (1 - e^{-2/5 \times 3})$$
$$= \cancel{1} - e^{-12/5} - \cancel{1} + e^{-6/5}$$
$$= e^{-6/5} - e^{-12/5}$$
$$= \underline{0.21047}$$



Q The time to failure of the chip follow exponential with a mean of 5000 hours. The chip is in operation for past 1000 hours. What is the probability that chip will be operating for another 6000 hours.

$E(x) = 1/\lambda = 5000 = 5$ in thousand hours

$\lambda = 0.0002 = \frac{1}{5000}$ $\quad 0.2$

$E(X) = \frac{1}{\lambda} \quad \lambda = \frac{1}{E(X)} = \cancel{\lambda} = \frac{1}{5000}$

$\therefore \cancel{P(x > 7000)} = 1 - P(X \leq 7000)$

$E(X) = \frac{1}{\lambda}$

$\therefore F(x) = 1 - e^{-\lambda x}, \quad x \geq 0$

$5 \text{ Thhrs.} = \frac{1}{\lambda}$

$\lambda = \frac{1}{5}$

$\therefore P(1 \leq x \leq 7) = F(7) - F(1)$

$= \cancel{1} - e^{-7/5} - \cancel{1} + e^{-1/5}$

$= 0.57213$

After 7000 hours of operations. What is the probab that it will work for 2000 hours more.

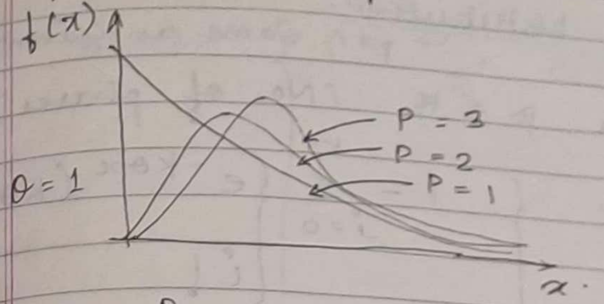
$\therefore P(7 \leq x \leq 9)$

$= F(9) - F(7)$

$= \cancel{1} - e^{-9/5} - \cancel{1} + e^{-7/5}$

$= 0.081298$

Gamma / Erlang Distribution



β = Shape parameter

θ = Scale parameter

pdf

$$F(x) = \begin{cases} \frac{\beta\theta}{\Gamma(\beta)} (\beta\theta)^{\beta-1} e^{-\beta\theta \cdot x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

$E(x) = \frac{1}{\theta}, \quad V(x) = \frac{1}{\beta\theta^2}$

cdf

$$F(x) = \begin{cases} 1 - \int_x^\infty \frac{\beta\theta}{\Gamma(\beta)} (\beta \cdot \theta t)^{\beta-1} e^{-\beta\theta t} \, dt & x > 0 \\ 0 & x \leq 0 \end{cases}$$

if β is an integer

$\Gamma(\beta) = (\beta - 1)!$

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Erlang Distribution.

→ pdf same as Gamma distribution.

when $\beta = k$ (No. of phases)

$$F(x) = \begin{cases} 1 - \sum_{i=0}^{k-1} \frac{e^{-k\theta x}(k\theta x)^i}{i!} & x > 0 \\ 0 & x \le 0 \end{cases}$$

cdf.

$$E(x) = \frac{1}{\theta}, \quad V(x) = \frac{1}{k\theta^2}$$

Q A medical examination is given in 3 stages by a physician. Each stage is exponentially distributed with a mean service time of service as 20 min. Find the probability that the exam will take 50 min or less. Compute expected time of exam.

$k = 3$.

$E(x) = 20$ min.

$\theta = \frac{1}{20}$ (for 1 stage)

for 3 stages $= \frac{1}{20 \times 3}$

$x = 50$ min.

$\theta = \frac{1}{60}$

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$$\therefore F(x) = 1 - \sum_{i=0}^{2} \frac{e^{-3(1/20\times3)50}(3/_{20\times3} \times 50)^i}{i!}$$

$$= 1 - \left[\frac{e^{-2.5}(2.5)^0}{0!} + \frac{e^{-2.5}(2.5)^1}{1!} + \frac{e^{-2.5}(2.5)^2}{2!}\right]$$

$= 1 - [0.54381]$

$= \underline{0.4561}$.

$\therefore \theta = 1/60$.

$\therefore E(x) = 60$ min $= (1/\theta)$.

⇓

is the expected time of exam.

Q A professor gives 6 ~~problems~~ students in each exam. Each exam requires average of 30 min grading time for 15 students. Grading time is exponential.

i) Find out probability that the professor will finish grading in $2\frac{1}{2}$ hours or less.

ii) What is the most likely grading time?

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$$\underset{x=2160}{F(x)} = 1 - \sum_{i=0}^{1} \frac{e^{-2 \frac{1}{2000} \times 2160} \left(-2 \frac{\times 2160}{2000}\right)^i}{i!}$$

$$= 1 - \left[\frac{e^{-2.16}(2.16)^0}{0!} + \frac{e^{-2.16}(2.16)^1}{1!} \right]$$

$$= 1 - e^{-2.16}\left[1 + 2.16\right]$$

$$= 1 - 0.3644$$

$$F(x) = \underline{0.6355}$$

Q Daily demand for rice at a departmental store in 1000s of kg is found to follow gamma distribution with shape parameter 3 and scale parameter $1/2$

Determine the probability of daily demand exceeding 5000 kg on a given day.

$\beta = 3$ $x = 5.000$

$\theta = 1/2$.

Gamma Distribution.

$$\therefore F(x) = 1 - \int_x^{\infty} \frac{\beta\theta}{\Gamma(\beta)} (\beta\theta t)^{\beta-1} e^{-\beta\theta t} d$$

By Erlang Distribution.

$K = \beta = 3$.

$x = 5.000$

$\theta = 1/2$ (given)

$$\underset{x=5000}{F(x)} = 1 - \sum_{i=0}^{2} \frac{e^{-7.500}(7.500)^i}{i!}$$

$$= 1 - \left[e^{-7.500} \left(\frac{(7.500)^0}{0!} + \frac{(7.500)^1}{1!} + \frac{(7.500)^2}{2!} \right) \right]$$

$= 1 -$ (28132501)

$= 1 - [0.020256]$

$= \underline{0.979743}$

17/8/2010

mid	(F) frequency	d	Fd	Fd^2	(F/1000) Probability
409.5 to 414.5 (412)	6	-4	-24	96	0.006
417	34	-3	-102	306	0.034
422	132	-2	-264	528	0.132
427	179	-1	-179	179	0.179
432	218	0	0	0	0.218
437	183	1	183	183	0.183
442	146	2	292	584	0.146
447	69	3	207	621	0.069
452	30	4	120	480	0.030
457	3	5	15	75	0.003
	1000	5	248	3052	

$\sigma \Rightarrow$ Standard Deviation

$$= I \times \sqrt{\frac{\Sigma Fd^2}{\Sigma F} - \left(\frac{\Sigma Fd}{\Sigma F}\right)^2}$$

$$= 5 \times \sqrt{\frac{3052}{1000} - \left(\frac{248}{1000}\right)^2}$$

$$= 5 \times \sqrt{3.052 - 0.061504}$$

$$= \underline{\underline{8.6465}} \approx \underline{\underline{9}}$$

(I) Interval: 5 (Difference).

A = 432

↳ Middle value.

$$\mu = \text{mean} = A + \frac{\Sigma Fd}{\Sigma F} \times I$$

$$\boxed{d = \frac{x - A}{5}}$$

$$= 432 + \frac{248}{1000} \times 5$$

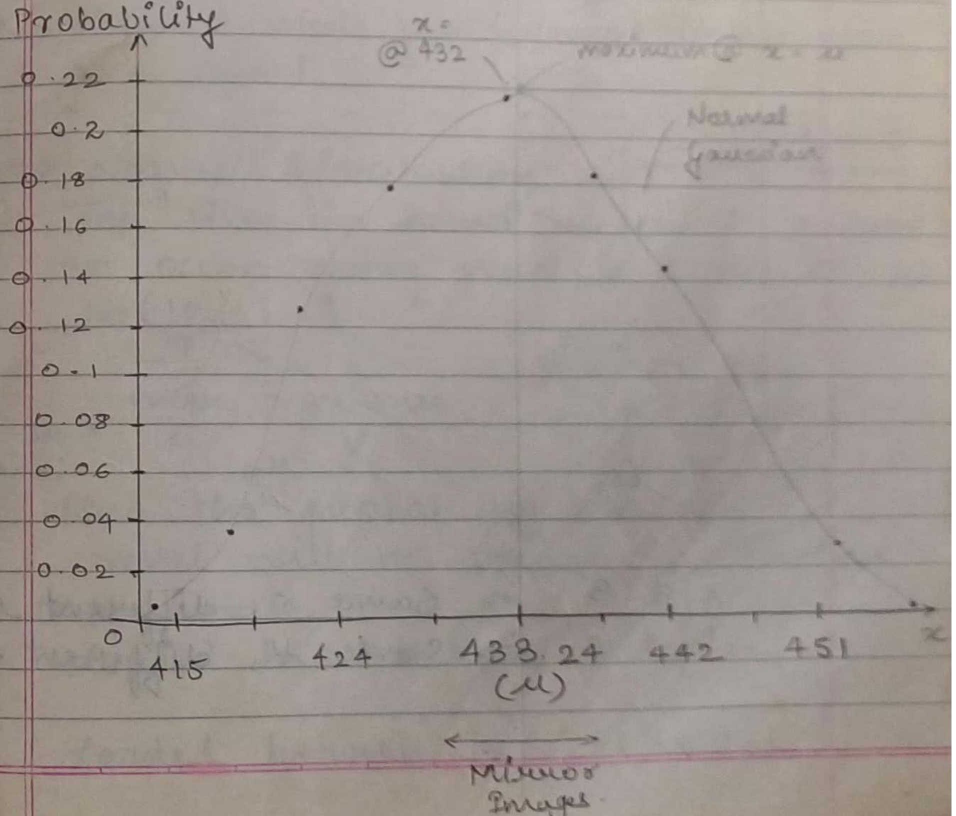
$$= 432 + (0.248 \times 5)$$

$$= 432 + 1.24$$

$$\underline{\mu} = \underline{433.24}$$

⇑

This is my actual mean

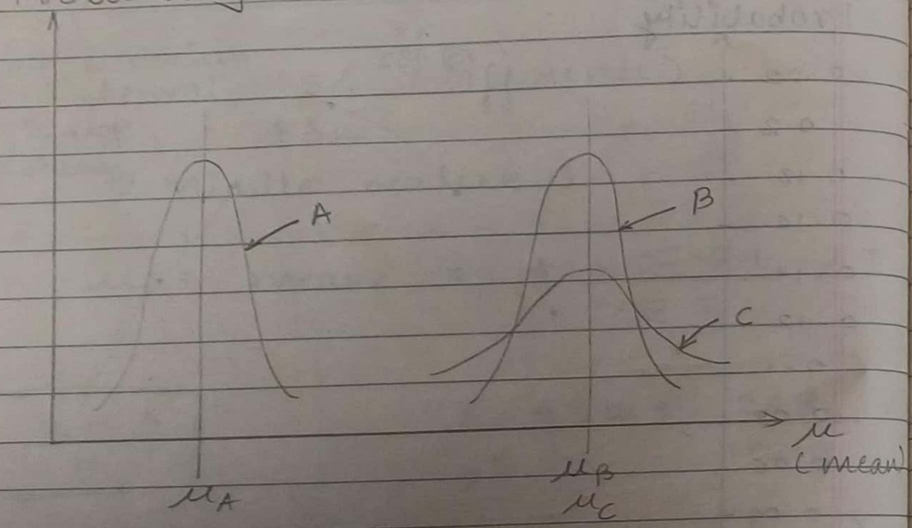


PDF:

$$\phi(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-z^2/2}$$

Z = Standard Normal Variate $\quad z = \frac{x-\mu}{\sigma}$

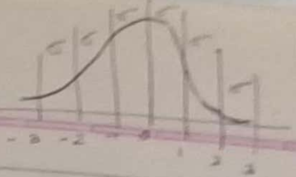
$$\phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\, dt$$



A & B => Same σ, different μ.

B & C => Same μ, different σ

6σ =



i) Each region (i.e. region to the left of the mean & region to the right of the mean) has 0.5 probability. As $x \to \infty$ or $x \to -\infty$ the value of probability approaches zero.

ii) $F(\mu - x) = F(\mu + x)$

i.e. PDF is symmetric about μ

iii) Maximum value of PDF occurs at $x = \mu$

iv) Mean & mode for normal distribution are equal.

20/8/2010

⇒ Normal Distribution:

The time in hours required to load an ocean going vessel is distributed as $N(12, 4)$

$\uparrow$ mean μ $\quad$ $\uparrow$ variance V

$\sqrt{V} = \sigma$

$\therefore \sigma = \sqrt{V}$

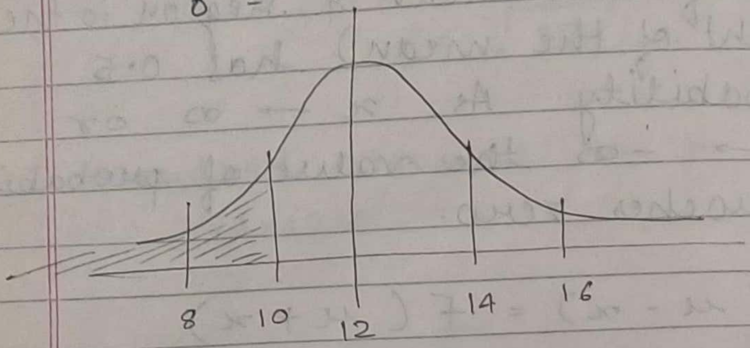
a) Find the probability that the vessel will be loaded in less than 10 ~~12~~ hours.

b) Loaded in ~~less than~~ ~~10~~ 12 or more hours.

c) Loaded between 10 to 12 hours.

a) $\mu = 12$.

$\sigma = 2$



CDF:

$$\int_{-\infty}^{10}$$

$$Z = \frac{x - \mu}{\sigma}$$

$$Z = \frac{10 - 12}{2} = -$$

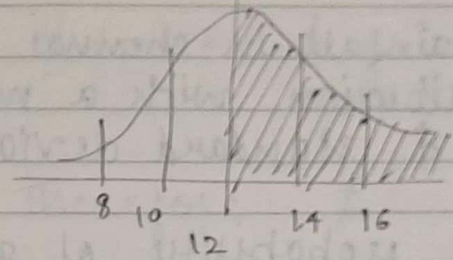
$$F(10) = \phi(10) \neq \left[\left(Z = \frac{10 - 12}{2} \right) \right.$$

$\therefore \phi(-1)$ from table

$$\phi(1) = 0.8413$$

$$\phi(-1) = 1 - \phi(1)$$
$$= 1 - 0.8413$$
$$= \underline{0.1587}$$

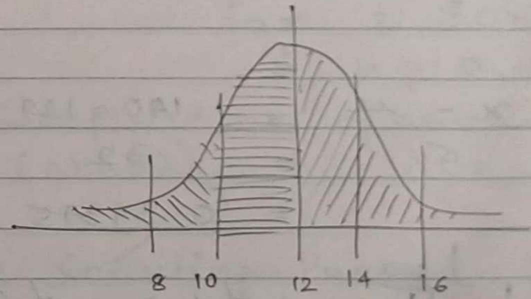
b).



By looking at graph

Probability $= \underline{0.5}$

c)



Probability:

$= 1 - P(>12) - P(<10)$

$= 1 - 0.5 - 0.1587$

$= \underline{0.3413}$.

Q. The annual rainfall in chennai is normally distribution with a mean of 129 cm & standard deviation of 32 cm.

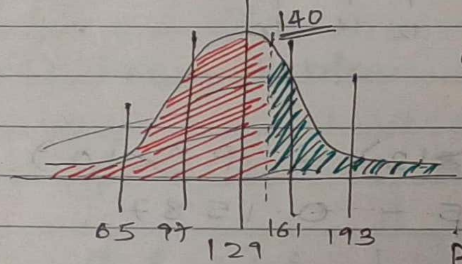
i> What is the probability of getting excess rain. >140

ii> Probability of getting deficient rain. <80 m

Ans: $\phi(z) =$

i> $z = \frac{x - \mu}{\sigma} = \frac{140 - 129}{32} = \frac{11}{32}$

$= 0.34375$



$\phi(z) = \phi(0.34375)$

$= 0.6331$

(which is for 140)

But we want >140

$\therefore \phi(z) = 1 - \phi(0.34375)$

$= 1 - 0.6331 = 0.366$

ii> $z = \frac{80 - 129}{32} = \frac{-49}{32}$

$\therefore z = (-1.53)$

$\phi(z) = \phi(1.53)$

$= 0.9367$

Therefore $\phi(-1.53) = 1 - 0.9367$

$= 0.0633$

23/8/2010

Poisson Process

probability

$P[N(t) = n)] = \frac{e^{-\lambda t} (\lambda t)^n}{n!}$

for $t \geq 0$

& $n = 0, 1, 2, \ldots$

expected no. of arrivals

$E(N(t)) = \alpha = \lambda t = \Lambda [N(t)]$

⇒ The Counting Process

→ $[N(t), t \geq 0]$ where $t \geq 0$.

The counting process is said to be a poisson process with mean rate 'λ', if:-

a> arrivals occur one at a time.

b> $[N(t) \quad t \geq 0]$ has stationary increments.

⇒ meaning:

The distribution of number of arrivals between time t & $t + s$ depents on the length of intervals s and not on starting point t.

i.e. Arrivals are completely random without rush or slag periods.

c> $[N(t), t \geq 0]$ has independent increments.

$\underbrace{\hspace{6cm}}$

Large or small number of arrivals in one time interval has no effect on ~~subsequent~~ number of arrivals in subsequent time intervals.

d> NSPP: Non-Stationary Poission Process.

from Poission process — If we drop assumption No. 2, we have a non-stationary poission process characterised by $\lambda(t)$ i.e the arrival rate at time 't'

$$\lambda(t) = \begin{cases} 30, & 0 \leq t < 1 \quad \text{6 am to 7 am} \\ 45, & 1 \leq t < 2 \quad \text{7 am to 8 am} \\ 20, & 2 \leq t < 4 \quad \text{8 am to 10 am} \end{cases}$$

Arrival rate of customers to a breakfast restaurant that is open from 6 to 9 am.

Assume NSPP model is appropriate

a> Derive $\Lambda(t)$

b> Compute the arrivals between 7.30 to 8.30 am

c> Compute the probability that there are fewer than 60 arrivals between 7.30 to 8.30 am.

$$\Lambda(t) = \int_0^t \lambda(s)\, ds$$

6	0
6 30	0.5
7	1
7 30	1.5
8.00	2
8 30	2.5
9	3

$$\lambda(t) = \begin{cases} \int_0^t 30\, ds = 30t & 0 \leq t < 1 \\ \int_0^1 30\, ds + \int_1^t 45\, ds = & 1 \leq t < 2 \\ \int_0^1 30 ds + \int_1^2 45 ds + \int_2^t 20 ds = & 2 \leq t < 4 \end{cases}$$

$$\Lambda(t) = \begin{cases} \int_0^t 30\, ds = 30t & 0 \le t < 1 \\ \int_0^1 30\, ds + \int_1^t 45\, ds = 30 + 45[t-1] = 45t - 15 & 1 \le t < 2 \\ \int_0^1 30\, ds + \int_1^2 45\, ds + \int_2^t 20\, ds = 30 + 45 + 20[t-2] & 2 \le t < 4 \end{cases}$$

$$= 75 + 20t - 40$$

$$= \underline{35 + 20t}$$

b) $\Lambda(2.5) - \Lambda(1.5)$.

$$= [20(2.5) + 35] - [45(1.5) - 15]$$

$$= (50 + 35) - (67.5 - 15)$$

$$= 85 - 52.5$$

$$= \underline{32.5}$$

$\lambda t \Rightarrow$ Expected no. of arrivals

c) $$P[(N(t) = n)] = \frac{e^{-\lambda t} (\lambda t)^n}{n!} \qquad t \ge 0$$

$$P[N(2.5) - N(1.5) = k]$$

$$= \frac{e^{-(\Lambda(2.5) - \Lambda(1.5))} (\Lambda(2.5) - \Lambda(1.5))^k}{k!}$$

$$= \frac{e^{-(32.5)} (32.5)^k}{k!}$$

$$= \sum_{k=0}^{59} \frac{e^{-32.5} (32.5)^k}{k!}$$

$$= e^{-32.5} \sum_{k=0}^{59} \frac{(32.5)^k}{k!}$$

$$= e^{-32.5} \left[1 + \frac{(32.5)^1}{1!} + \frac{(32.5)^{59}}{59!} \right]$$

Q2 Arrivals at a post office occur at a rate of 2 per minute from 8 am to 12 pm, then drop to 1 every 2 minutes until the day ends at 4 pm. What is the probability distribution of the no. of arrivals between 11 am to 2 pm.

Ans.

$$\lambda(t) = \begin{cases} 2 & 0 \le t < 4 \quad (8\text{ am to } 12) \\ 1/2 & 4 \le t < 8 \quad (12\text{ pm to } 4\text{ pm}) \end{cases}$$

$$\Lambda(t) = \begin{cases} \int_0^t 2\,ds = 2t & 0 \le t < 4 \\ \int_0^4 2\,ds + \int_4^t 1/2\,ds = 8 + \frac{1}{2}\Big|_4^t = \frac{1}{2}t + 6 & 4 \le t < 8 \end{cases}$$

From 11 to 2 pm.

$\Lambda(6) - \Lambda(3)$.

$=$

$= 3 + 6 - 6 = 3.$

$$P[(N(6) - N(3)) = k]$$

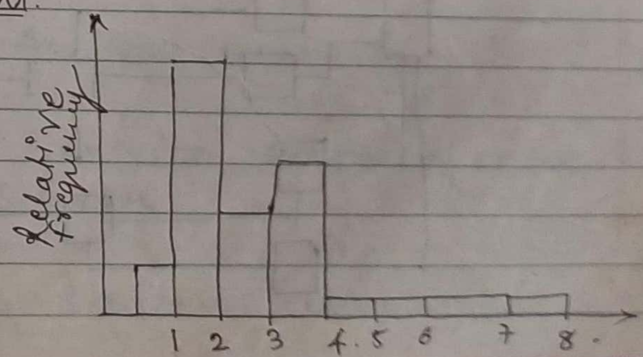
$$= \frac{e^{-3}(3)^k}{k!} \Rightarrow \text{is the distribution}$$

Empirical Distribution.

Arrivals	Frequency	Relative frequency	Cumulative Frequency
1	30	30/T = 0.1	0.1
2	110	0.37	0.47
3	45	0.15	0.62
4	71	0.24	0.86
5	12	0.04	0.9
6	13	0.04	0.94
7	7	0.02	0.96
8.	12	0.04	1.0

Total = 300.

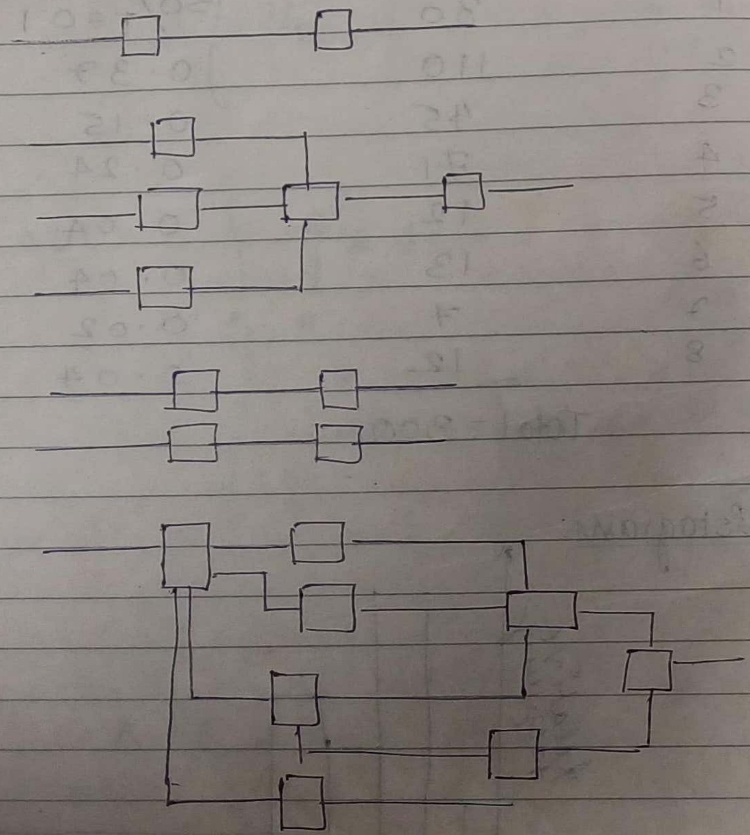
Histogram:



26/8/2010

Queuing Theory:

→ Calling populator.
→ System capacity.
→ Queue Behaviour & Discipline.
→ Arrival Process.
→ Service Time & Service Mechanism
→ Queuing Notation
(Kendall's Notation)



Queuing Notation.

Kendall's Notation A / B / c / N / K

small c

Steady state parameters

A represents → Inter-arrival time distribution

B —— " —— → Service time distributions

c —— " —— → No. of parallel servers.

K —— " —— → Size of calling population.

N —— " —— → System capacity

P_n = Steady state probability of having n customers in the system

$P_n(t)$ = Probability of n customers in system at time t

λ => Arrival rate.

μ => Service rate of one server.

ρ => Server utilization

W_n = Total time spent in the system by n^{th} arriving customer.

W_n^Q = Total time spent in waiting line by the customer n

$L(t)$ = No. of customers in the system at time t.

$L_Q(t)$ = No. of customers in the queue at time t.

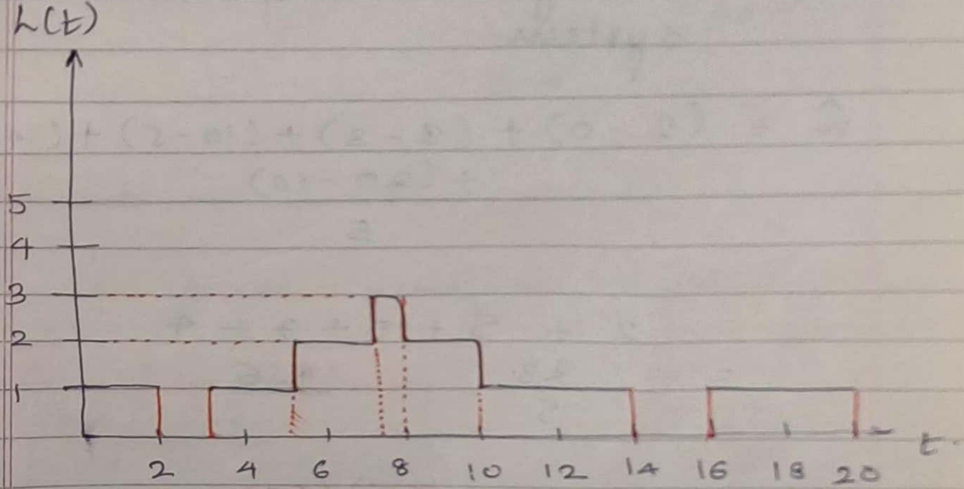
L = Long run time average number of customer in the system.

L_Q = Long run time average number of customers in the queue.

W = Long run average time spent in the system per customer.

W_Q = Long run average time spent in the queue per customer.

Ex: Arrival - 0, 3, 5, 7, 16
Departure - 2, 8, 10, 14, 20



Time weighted Average of number of people:

$$\hat{L} = \frac{1}{T}\sum_{i=0}^{\infty} i T_i$$

0 people in system for 3 time units.

$$\hat{L} = \frac{1}{20}\left(0(3) + \underbrace{1(12)}_{} + 2(4) + 3(1)\right)$$

12 time units with 1 people.

$$= \frac{1}{20}\left[12 + 8 + 3\right]$$

$$= \frac{23}{20}$$

$\hat{L} = 1.15$

$\hat{W}$ = Average time spent in the system

$$\hat{W} = \frac{(2-0)+(8-3)+(10-5)+(14-7)+(20-16)}{5}$$

$= 2 + 5 + 5 + 7 + 4$

$= \frac{23}{5} = 4.6$

$$\hat{W}_q = \frac{0 + 0 + \overset{(8-5)}{3} + \overset{(10-7)}{3} + 0}{5}$$

$= \frac{6}{5} = 1.2$

zero people in queue for 15 time units.

$$\hat{L}_q = \frac{0(15) + 1(4) + 2(1)}{20}$$

two people in queue for 1 time unit.

$= \frac{4+2}{20} = \frac{6}{20} = 0.3$

27/8/2010

Little's Law: $L = \lambda W$.

$L = \frac{5}{20} \cdot 4.6$ (for previous example)

$= \frac{1}{4} \cdot \cancel{4.6}$ 1.15

$L = 1.15$

$\therefore$ L is verified from Little's Law.

Queuing Systems:

M/G/1

G → Random Service Times

$$\rho = \frac{\lambda}{\mu}$$

$$L = \rho + \frac{\lambda^2 (1/\mu^2 + \sigma^2)}{2(1-\rho)}$$

$$W = \frac{1}{\mu} + \frac{\lambda(1/\mu^2 + \sigma^2)}{2(1-\rho)}$$

$$L_q = \frac{\lambda^2 (1/\mu^2 + \sigma^2)}{2(1-\rho)}$$

$$P_0 = 1 - \rho$$

M/M/1.

$$\rho = \frac{\lambda}{\mu}$$

$$L = \frac{\lambda}{\mu - \lambda} = \frac{\rho}{1-\rho}$$

$$W = \frac{1}{\mu - \lambda} = \frac{1}{\mu(1-\rho)}$$

$$W_q = \frac{\lambda}{\mu(\mu-\lambda)} = \frac{\rho}{\mu(1-\rho)}$$

$$L_q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{\rho^2}{1-\rho}$$

$$P_n = (1-\rho)\rho^n$$

Arrival pattern => Exponential / Poission } M/M

Q) Malfunctions occur in a machine shop according to poission process. $\lambda = 1.5$ per hour. Repair times of a single mechanic take average 30 mins, with standard deviation of 20 mins. Find out steady state average number of broken machines.

Sol^n: $\lambda = 1.5$ per hour

mean service time (μ) & (standard Devial) are same then M/M/1

But here not same so we use M/G/1.

$\lambda = 1.5$ per/hour.

$\mu = 30$ min =

30 min → 1 machine

1 hr → 2 machines.

$\therefore \mu = 2$ per hour.

$\therefore \sigma = 20$ min.

$= 1/3$ hour.

$$\therefore L = \rho + \frac{\lambda^2 (1/\mu^2 + \sigma^2)}{2(1-\rho)}$$

$$\rho = \frac{\lambda}{\mu} = 0.75$$

$$L = 0.75 + \frac{1.5^2 (1/4 + 1/9)}{2(0.25)}$$

$$= \underline{2.375}$$

Q. Able & Baker are competing for a job. Able's average service time is faster than Baker's But Baker is more consistant. Arrivals arrive according to poission at $\lambda = 2$ hour.

Able has average service time of 24 min with standard deviation of 20 min.

Baker has average service tim of 25 min with standard deviation of 2 min.

If average length of the queue is critchia for hiring, which worker should be hired.

Q Model :- M/G/1

$$\lambda = \frac{1}{30} \text{ per min}$$

$$\mu_A = 24 \text{ min}$$

$$\sigma_A = 20 \text{ min}$$

$$\mu_B = 25 \text{ min}$$

$$\sigma_B = 2 \text{ min}$$

$$L_Q = \lambda^2 \frac{(1/\mu^2 + \sigma^2)}{2(1-\rho)}$$

$$\rho = \frac{\lambda}{\mu}$$

$$\rho_A = \frac{\lambda}{\mu_A} = \frac{1/30}{24} = \frac{1}{720}$$

$$\rho_B = \frac{\lambda}{\mu_B} = \frac{1/30}{25} = \frac{1}{750}$$

$$L_{Q(ABLE)} = (1/30)^2 \frac{(1/\mu_A^2 + \sigma_A^2)}{2(1-\rho_A)}$$

$$= (1/30)^2 \frac{(1/24^2 + 20^2)}{2(1 - 1/720)}$$

$$= \underline{0.2225}$$

Service time random => M/G/1
Service time exponential => M/M/1

$$L_{Q(BAKER)} = \frac{(1/30)^2 \left(1/\mu_B^2 + \sigma_B^2\right)}{2(1-\rho_B)}$$

$$= \frac{(1/30)^2 \left(1/25^2 + 2^2\right)}{2(1 - 1/750)}$$

$$= \underline{0.0022}$$

$\therefore L_{Q(B)} < L_{Q(A)}$

$\therefore$ Baker is to be hired.

Q <u>30/8/2010</u>

Q i) An airport is being constructed. Time to land an airplane is exponentially distributed with a mean time of 3/2 min. If arrivals are assumed to occur at random, what arrival rate can be tolerated if average wait in the sky is not to exceed 3 minutes.

2

Ans: $\mu = 1 / \frac{3}{2}$ min $= \frac{2}{3}$ min

Aververage service ~~time~~ rate $= \frac{1}{\text{Service time}}$

M/M/1

$W_Q \le 3$ min.

$$W_Q = \frac{\lambda}{\mu(\mu-\lambda)}$$

$$3 \ge \frac{\lambda}{\frac{2}{3}\left(\frac{2}{3} - \lambda\right)}$$

$\therefore \lambda \le 0.444$ / min.

Q ii) Patients arrive according to poission process at the rate of one per hour. The medical examination requires 3 stages each one independently exponentially distributed with a mean service time of 15 mins. A patient must go through all three stages before the next patient is admitted. Compute average number of delayed patients

6

$\lambda = 1$ / hour.

Service time is exponentially distributed for each stage But combined the service time is not exponentialy distributed.
So we choose M/G/1

$\mu = 1/45$ min for 3 stages.

Service rate = $\frac{1}{\text{Service time}}$

$\mu = 1/45$

$\sigma^2 = \sigma_1^2 + \sigma_2^2 + \sigma_3^2$

$\sigma^2 = 15^2 + 15^2 + 15^2$

$\lambda = 1/60$

$\mu = 1/45$

$$L_Q = \frac{(1/60)^2 \left(1/(1/45)^2 + (15^2 \times 3) \right)}{2(1 - 45/60)}$$

$= \underline{1.5}$

Q (6) Arrivals of customers with a mean of 45 per hour and service time is exponential with a mean of one minute.

Determine the following:

i> Probability of having 0, 5, 10, customers in the system.

ii> L_Q, L, W_Q, W

Ans:

M/M/1 bcoz service is exponential.

$\lambda = 45$/hour.

Service time = 1 minute.

Service rate = $\frac{1}{\text{Service time}} = \frac{1}{1/60}$ = 60 per hour.

$\mu = 60$ per hour.

$\rho = \frac{\lambda}{\mu} = \frac{45}{60} = \underline{\underline{\frac{3}{4}}}$

i> $P_n = (1-\rho)\rho^n$

$= (1 - 3/4)(3/4)^n$

$n = 0, 5, 10.$

$n = 0.$

$P_0 = (1 - 3/4)(3/4)^0.$

$= 1/4.$

$n = 5.$

$P_5 = (1 - 3/4)(3/4)^5$

$= \underline{0.05932}$

$n = 10$

$P_{10} = (1 - 3/4)(3/4)^{10}.$

$= \underline{0.01407}$

ii> $L_Q = \frac{\rho^2}{1-\rho} = \frac{(3/4)^2}{1/4}.$

$= \underline{2.25}$

$L = \frac{\rho}{1-\rho} = \underline{3}.$

DATE: / /

$$W_q = \frac{\rho}{\mu(1-\rho)} = \underline{0.05}$$

$$W = \frac{1}{\mu(1-\rho)} = \underline{\underline{0.06667}}$$

Q3 a) A machine shop repairs electric motors which arrive according to poission at the rate of 12 per week. (5 day week, 40 working hours) past data indicates that motors can be repaired on an average in 2.5 hours with a variance of $1(\text{hour})^2$. How many working hours a customer should expect to leave a motor at a shop.

b) If the variance of repair time could be controlled, what variance would reduce expected waiting time to 6.5 hours!

Ans: MG1

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$\lambda = 12$ per week

$\lambda = \frac{12}{40}$ per hour

Service time = 2.5

Service rate $(\mu) = \frac{1}{\text{Service time}}$

$\mu = \underline{0.4}$

$\underbrace{\sigma = \sqrt{V(x)}}_{\text{Poission Distributn}} = 1\text{ hr}$

$\therefore V(x) = \sigma^2 = 1\text{ hr}^2$

i) $W = \frac{1}{\mu} + \frac{\lambda(1/\mu^2 + \sigma^2)}{2(1-\rho)}$

$=$

$= \frac{1}{0.4} + \frac{12/40(1/0.4^2 + 1^2)}{2(1 - 12/40\times0.4)}$

$= 2.5 + 4.35$

$= \underline{6.85}$

ii) $W = 6.5$

$6.5 = \frac{1}{0.4} + \frac{12/40(1/0.4^2 + \sigma^2)}{2(1 - 12/40\times0.4)}$

$0.5\times4 = 0.3(6.25 + \sigma^2)$

$6.6667 - 6.25 = \sigma^2$

$\therefore V(x) = \sigma^2 = 0.416$

Q. 5 Trucks arrive according to poission. with average rate of 36 per hour. The unloading service is exponentially distributed with service rate of 39 truck per hour. Drivers make Rs 9/- per hour and do not unload the trucks. How much expense is incurred by the trucking company for idle time on part of each driver.

Ans:

M/M/1:

$\lambda = 36$ / hour.

$\mu = 39$ / hour.

$W \rightarrow$ ~~total time spent by customer~~ in system.

$$= \frac{1}{\mu - \lambda} = \frac{1}{39-36} = \frac{1}{3}$$

$= 0.333$ hr.

$\therefore$ Amount spent $= W \times 9$

$= 0.333 \times 9$

$= \frac{1}{\cancel{3}} \times \cancel{9}\,3$

$=$ Rs $\underline{3/-}$

M/G/∞

$P_0 = e^{-\lambda/w}$

$w = \frac{1}{\mu}$

$W_q = 0$

$L = \frac{\lambda}{\mu}$

$L_q = 0$

$$P_n = \frac{e^{-\lambda/w} (\lambda/\mu)^n}{n!} \quad n = 0, 1, 2, \ldots$$

Q 1) A large consumer shopping mall is being constructed. At busy times, the arrival rate of cars is expected to be 1000/hr. Studies indicate that consumers spend 3 hours for shopping. Mall designers want sufficient parking space for 99.99% of times. How many parking spaces should the mall have.

M/G/∞ since probability given is 99.99%.

$\lambda = 1000$/hr.

$\mu = 1/3$

$$P_n = \frac{e^{-\lambda/\mu} (\lambda/\mu)^n}{n!}$$

$$99.99 = \frac{e^{-3000} (3000)^n}{n!}$$

$$99.99 \times n! = e^{-3000} (3000)^n$$

Assume $n = 3000$

$$99.99 = \frac{e^{-3000} (3000)^n}{n!}$$

$$= \frac{e^{-3000} \times (3000)^{3000}}{3000!}$$

$$\log 99.99 = \log e^{-3000} + 3000 \log 3000 - \log(3000!)$$

$$= -3000 \log e + 3000 \log 3000 - \log(3000!)$$

$$= -1302.88344$$

$$+ 10431.36376$$

$-$

Random Number Generation

Random
Pseudo Number generation

→ Uniformly generated
→ Continuous valued
→ Mean of generated random number
→ Variance of generated random numbers
→ Independence of Random Numbers
=> Auto co-relation
=> Successively high or low values
=> Several numbers above/below mean are not allowed

Q $x_{i+1} = (a x_i + c) \bmod m$

$i = 0, 1, 2, \ldots$

x_0 = seed

a = multiplier

c = increment

m = modulus

when $c \neq 0$ → mixed C.M
Mixed Congruential Method

when $c = 0$ → multiplicative CM
Multiplicative congruential Method

i) $x_0 = 27$, $a = 17$